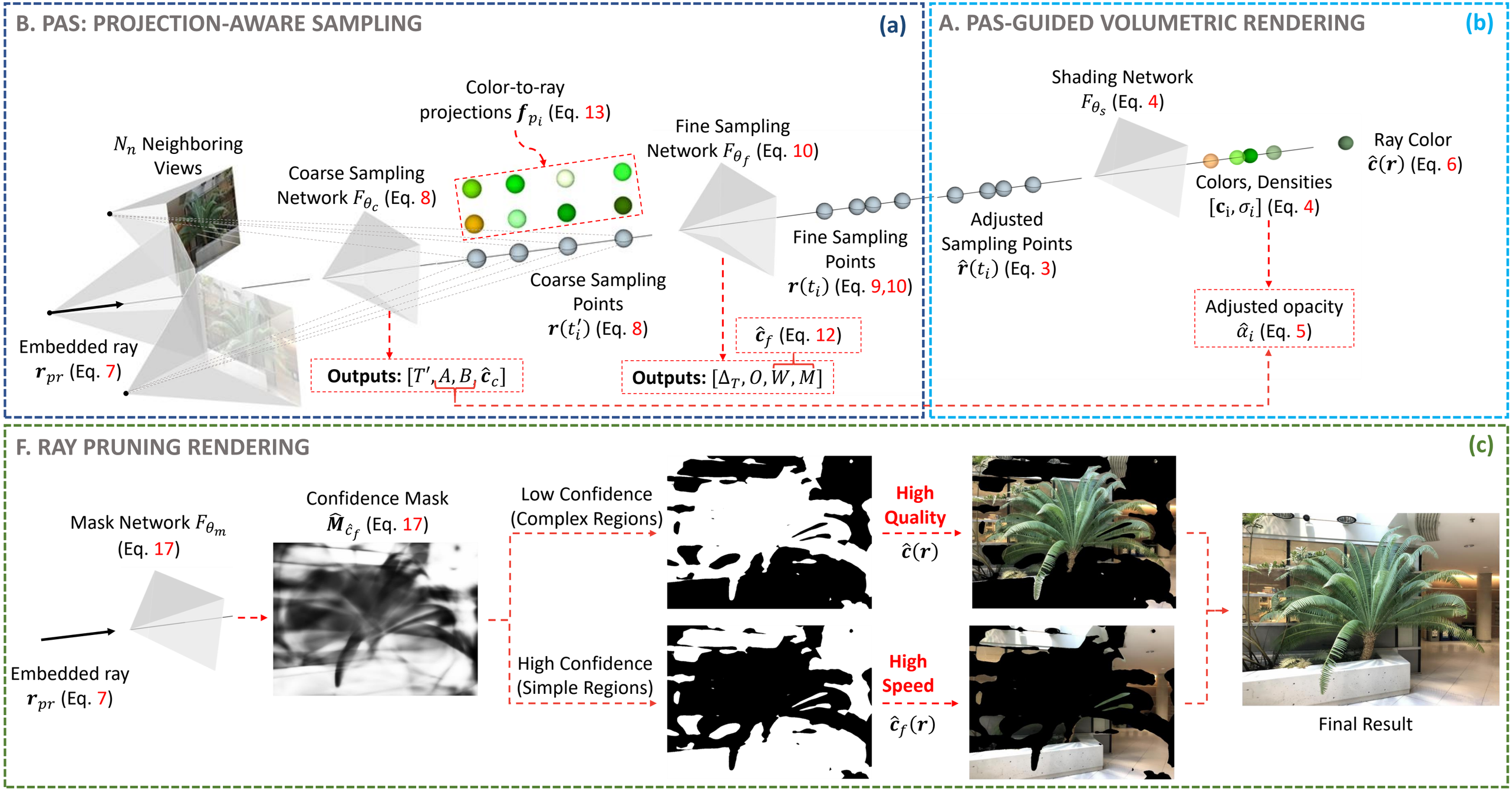
B. PAS: PROJECTION-AWARE SAMPLING
(a)
A. PAS-GUIDED VOLUMETRIC RENDERING
(b)
N_n Neighboring Views
Color-to-ray projections $\boldsymbol{f}_{p_i}$ (Eq. 13)
Coarse Sampling Network F_{θ_c} (Eq. 8)
Fine Sampling Network F_{θ_f} (Eq. 10)
Shading Network F_{θ_s} (Eq. 4)
Ray Color $\hat{\boldsymbol{c}}(\boldsymbol{r})$ (Eq. 6)
Colors, Densities $[\mathbf{c}_i, \sigma_i]$ (Eq. 4)
Adjusted Sampling Points $\hat{\boldsymbol{r}}(t_i)$ (Eq. 3)
Fine Sampling Points $\boldsymbol{r}(t_i)$ (Eq. 9,10)
Coarse Sampling Points $\boldsymbol{r}(t_i')$ (Eq. 8)
Embedded ray $\boldsymbol{r}_{pr}$ (Eq. 7)
Outputs: $[T', A, B, \hat{\boldsymbol{c}}_c]$
$\hat{\boldsymbol{c}}_f$ (Eq. 12)
Outputs: $[\Delta_T, O, W, M]$
Adjusted opacity $\hat{\alpha}_i$ (Eq. 5)
F. RAY PRUNING RENDERING
(c)
Mask Network F_{θ_m} (Eq. 17)
Confidence Mask $\widehat{\boldsymbol{M}}_{\hat{c}_f}$ (Eq. 17)
Low Confidence (Complex Regions)
High Confidence (Simple Regions)
High Quality
$\hat{\boldsymbol{c}}(\boldsymbol{r})$
High Speed
$\hat{\boldsymbol{c}}_f(\boldsymbol{r})$
Embedded ray $\boldsymbol{r}_{pr}$ (Eq. 7)
Final Result